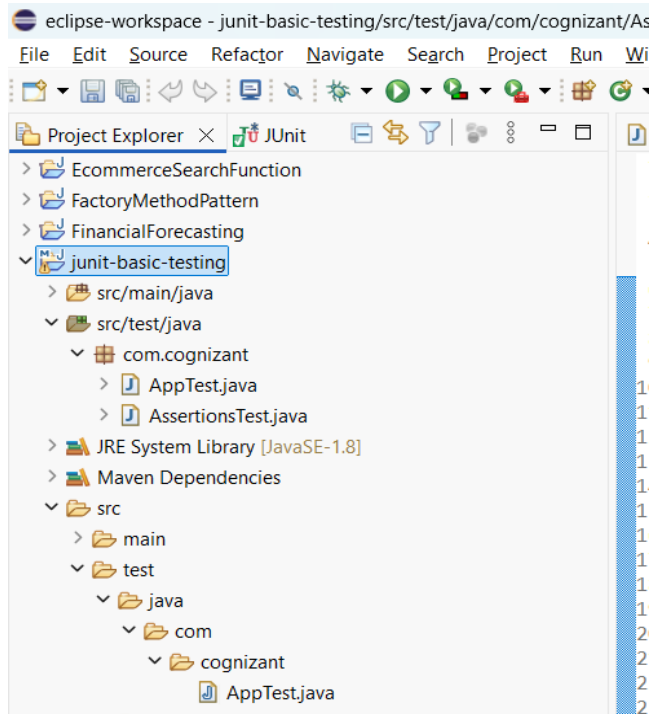


JUnit Testing Exercises

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

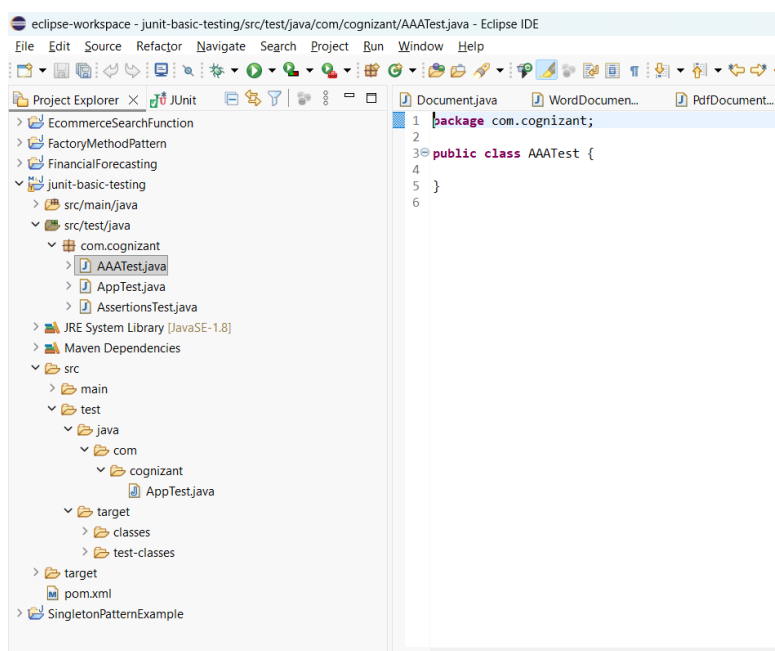
Step 1: Open the Existing Maven Project

Opened the existing Maven project junit-basic-testing in Eclipse IDE.



Step 2: Create a New Test Class

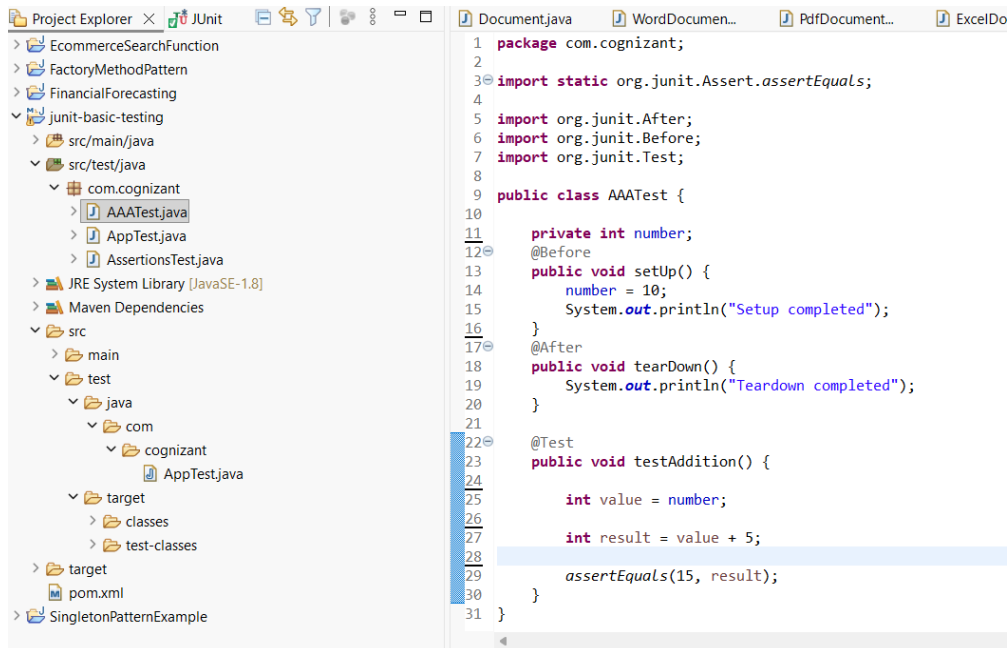
Right-click on the com.cognizant package inside src/test/java and create a new Java class named AAATest.



```
1 package com.cognizant;  
2  
3 public class AAATest {  
4  
5 }  
6
```

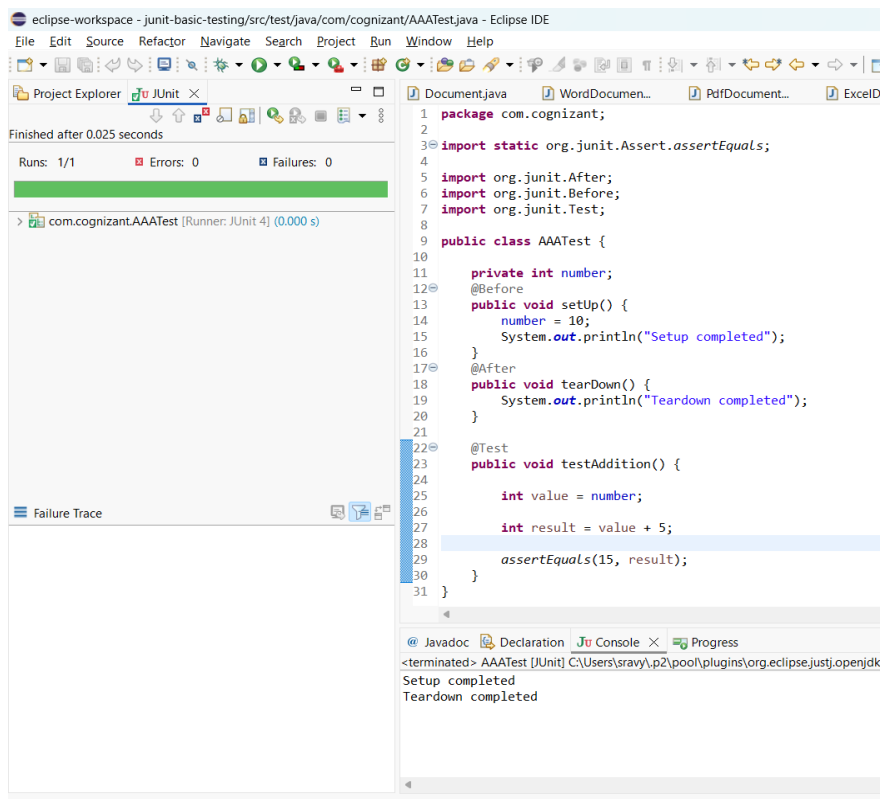
Step 3: Write the JUnit Program

Implemented the Arrange-Act-Assert (AAA) pattern using the `@Before`, `@Test`, and `@After` annotations.



Step 4: Run the JUnit Test

Executed the `AAATest` class using Run As → JUnit Test.



Step 5: Output

The JUnit test executed successfully.

Output:

Runs : 1/1

Errors : 0

Failures : 0

Setup completed

Teardown completed

Result

The Arrange-Act-Assert (AAA) pattern was implemented successfully using JUnit. The **@Before** method initialized the test data, the **@Test** method verified the expected result, and the **@After** method performed cleanup after execution. The test ran successfully with **1 test executed, 0 errors, and 0 failures**.

Teardown Methods in JUnit

Teardown methods are used to perform cleanup operations after the execution of each test case. They are annotated with **@After** in JUnit 4. These methods release resources, close database connections, clear temporary data, or reset variables so that the next test starts with a clean environment.

Example:

```
@After
public void tearDown() {
    System.out.println("Teardown completed");
}
```

Output:

Teardown completed